Clustering Analysis Documentation

Project: Customer Churn Analysis for Telecommunications Company
Team: Team 3
Stage: Stage 2 – Data Preparation and Clustering Analysis
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1. Purpose of Clustering Analysis

The purpose of the clustering analysis was to identify distinct customer segments within the telecommunications customer dataset based on customer characteristics.

K-Means clustering was used as an unsupervised machine learning technique to group customers with similar characteristics. The resulting customer segments can assist with understanding different customer groups and identifying opportunities for targeted retention and customer engagement strategies.

The clustering analysis was performed using the prepared and scaled dataset produced during the data preparation stage.

2. Clustering Dataset

The clustering-ready dataset used for the analysis was:

`ATS_cluster_ready_v1.csv`

The dataset contained:

- **6,741 customer records**
- **10 clustering features**
- **0 missing values**

The features used for clustering were:

1. `gender`
2. `SeniorCitizen`
3. `Dependents`
4. `tenure`

5. PhoneService
6. MultipleLines
7. MonthlyCharges
8. InternetService_Fiber optic
9. Contract_One year
10. Contract_Two year

The Churn target variable was **not included as an input feature** in the K-Means clustering process. This allowed customer groups to be identified from their characteristics rather than their known churn outcome.



3. Identifying the Optimal Number of Clusters

The **Elbow Method** was used to help identify an appropriate number of clusters.

K-Means models were evaluated for values of K from **1 to 10**. For each value of K, the within-cluster sum of squares (inertia) was calculated.

As the number of clusters increases, inertia decreases because customers can be assigned to more specific groups. The Elbow Method looks for the point where adding additional clusters produces progressively smaller improvements.

The analysis showed a noticeable change in the rate of improvement at approximately **K = 4**. Therefore, four clusters were selected for the final customer segmentation.

```python
inertias = []
k_values = range(1, 11)

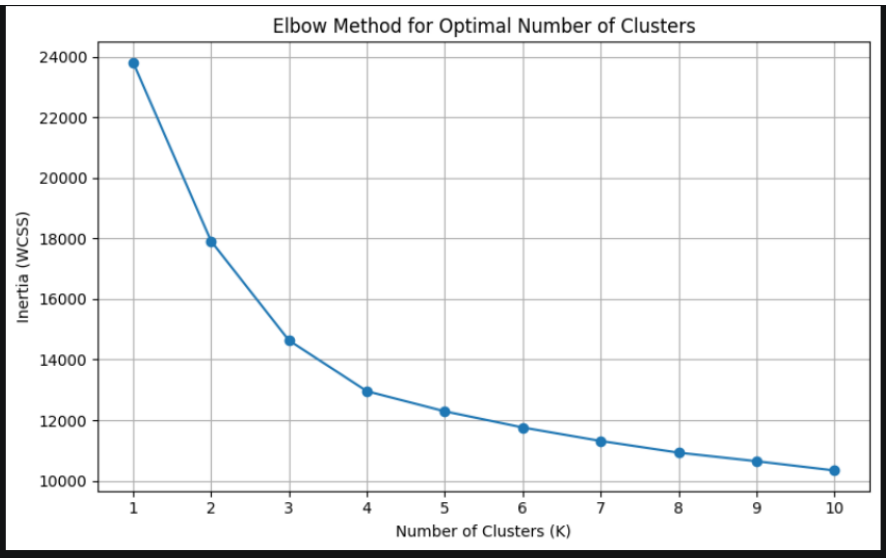
for k in k_values:
    kmeans_test = KMeans(
        n_clusters=k,
        random_state=RANDOM_STATE,
        n_init=10
    )

    kmeans_test.fit(X_cluster)
    inertias.append(kmeans_test.inertia_)

plt.figure(figsize=(8, 5))
plt.plot(k_values, inertias, marker="o")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Inertia (WCSS)")
plt.title("Elbow Method for Optimal Number of Clusters")
plt.xticks(k_values)
plt.grid(True)
plt.tight_layout()

plt.savefig("elbow_method.png", dpi=300, bbox_inches="tight")
plt.show()

for k, inertia in zip(k_values, inertias):
    print(f"K = {k}: Inertia = {inertia:.2f}")
```




```
K = 1: Inertia = 23823.26
K = 2: Inertia = 17918.02
K = 3: Inertia = 14634.40
K = 4: Inertia = 12956.38
K = 5: Inertia = 12291.47
K = 6: Inertia = 11757.57
K = 7: Inertia = 11312.20
K = 8: Inertia = 10925.12
K = 9: Inertia = 10643.53
K = 10: Inertia = 10336.91
```

Interpretation: The Elbow Method shows a noticeable reduction in the rate of improvement after approximately K=4. Based on this result, four clusters were selected for the final K-Means analysis.

4. Silhouette Analysis

Silhouette scores were also calculated as a secondary measure to understand the separation and cohesion of the potential cluster solutions.

The results included:

- K=2: **0.2233**
- K=3: **0.2134**
- K=4: **0.1998**

The highest silhouette score occurred at K=2. However, the Stage 2 analysis specifically required identification of the optimal number of clusters using the **Elbow Method**, which showed a meaningful bend around K=4.

K=4 was therefore selected because it provided a more detailed customer segmentation while remaining consistent with the Elbow Method analysis.



```
silhouette_scores = []
silhouette_k_values = range(2, 11)

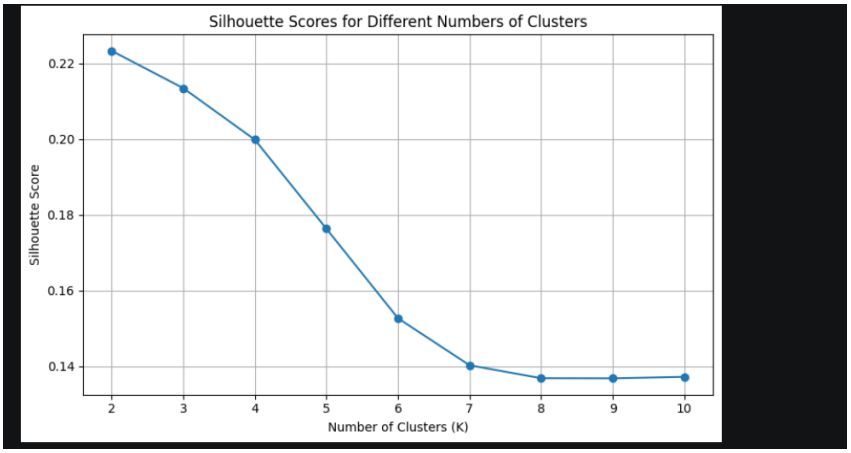
for k in silhouette_k_values:
    kmeans_test = KMeans(
        n_clusters=k,
        random_state=RANDOM_STATE,
        n_init=10
    )

    labels = kmeans_test.fit_predict(X_cluster)
    score = silhouette_score(X_cluster, labels)
    silhouette_scores.append(score)

plt.figure(figsize=(8, 5))
plt.plot(silhouette_k_values, silhouette_scores, marker="o")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Silhouette Score")
plt.title("Silhouette Scores for Different Numbers of Clusters")
plt.xticks(silhouette_k_values)
plt.grid(True)
plt.tight_layout()

plt.savefig("silhouette_scores.png", dpi=300, bbox_inches="tight")
plt.show()

for k, score in zip(silhouette_k_values, silhouette_scores):
    print(f"K = {k}: Silhouette Score = {score:.4f}")
```



This section is useful because it transparently shows that we considered another validation measure rather than incorrectly claiming that silhouette score selected K=4.

5. K-Means Model Training

Following the Elbow Method analysis, the final K-Means model was trained using **four clusters**.

The model used:

- **Number of clusters:** 4
- **Random state:** 42
- **n_init:** 10

The K-Means model assigned each of the **6,741 customers** to one of four clusters.

```
optimal_k = 4

kmeans = KMeans(
    n_clusters=optimal_k,
    random_state=RANDOM_STATE,
    n_init=10
)

cluster_labels = kmeans.fit_predict(X_cluster)

print("K-Means model trained successfully.")
print("Selected number of clusters:", optimal_k)

print("\nNumber of customers in each cluster:")
cluster_counts = pd.Series(cluster_labels).value_counts().sort_index()

for cluster, count in cluster_counts.items():
    print(f"Cluster {cluster}: {count} customers")
```

0.1s

```
K-Means model trained successfully.
Selected number of clusters: 4

Number of customers in each cluster:
Cluster 0: 1586 customers
Cluster 1: 1882 customers
Cluster 2: 1123 customers
Cluster 3: 2150 customers
```

The trained model was saved as:

`trained_kmeans_model.pkl`

This model file is included in the `Clustering_Analysis` folder.

6. Cluster Profiling and Labelling

After K-Means assigned customers to clusters, the resulting groups were profiled using customer characteristics and observed churn outcomes.

The four clusters were labelled as follows:

Cluster 0 – Newer, Lower-Spend Customers

- Customers: **1,586**
- Average tenure: **11.24 months**
- Average Monthly Charges: **$33.80**
- Observed churn rate: **24.91%**

This segment contains customers with relatively short tenure and lower monthly charges.

Cluster 1 – Long-Term, High-Value Customers

- Customers: **1,882**
- Average tenure: **58.73 months**
- Average Monthly Charges: **$93.17**
- Observed churn rate: **15.62%**

This group contains longer-term customers with relatively high monthly charges.

Cluster 2 – Long-Term, Low-Risk Customers

- Customers: **1,123**
- Average tenure: **54.35 months**
- Average Monthly Charges: **$34.90**
- Observed churn rate: **4.81%**

This segment contains longer-term customers with lower monthly charges and the lowest observed churn rate among the four clusters.

Cluster 3 – Newer, High-Risk Customers

- Customers: **2,150**
- Average tenure: **15.21 months**
- Average Monthly Charges: **$81.72**
- Observed churn rate: **48.74%**

This group contains relatively newer customers with higher monthly charges and the highest observed churn rate among the four segments.

7. Cluster Visualisation

The resulting customer clusters were visualised using **tenure** and **MonthlyCharges**.



The visualisation demonstrates differences between customer groups based on customer tenure and monthly charges.

The visualisation indicates that:

- Cluster 0 mainly represents customers with shorter tenure and lower monthly charges.
- Cluster 1 mainly represents customers with longer tenure and higher monthly charges.
- Cluster 2 mainly represents customers with longer tenure and lower monthly charges.
- Cluster 3 mainly represents customers with shorter tenure and higher monthly charges.

The K-Means model itself was trained using all **10 clustering features**. Therefore, this two-dimensional visualisation using tenure and MonthlyCharges is intended to provide an interpretable view of the resulting clusters rather than represent every feature used by the model.

8. Business Insights and Actionable Recommendations

The clustering analysis provides several useful customer insights.

Cluster 0 – Newer, Lower-Spend Customers:
These customers may benefit from early engagement, onboarding support and relevant targeted offers to encourage longer-term retention.

Cluster 1 – Long-Term, High-Value Customers:
These longer-term, higher-spending customers may benefit from loyalty initiatives, personalised service, retention benefits and relevant value-added offers.

Cluster 2 – Long-Term, Low-Risk Customers:
This group has the lowest observed churn rate. Maintaining customer satisfaction while considering suitable cross-selling or service upgrade opportunities may be appropriate for this segment.

Cluster 3 – Newer, High-Risk Customers:
This segment has the highest observed churn rate at **48.74%**. Retention efforts could focus on early customer check-ins, service reviews, targeted offers and further investigation into customer value perception.

These recommendations are based on descriptive patterns identified through the clustering analysis and observed churn outcomes. They should not be interpreted as evidence that any individual customer characteristic directly causes churn.

9. Output Files

The following clustering analysis files were produced:

- `Rupshana_KMeans_Clustering_ATS.ipynb` – complete clustering analysis and Python code.
- `trained_kmeans_model.pkl` – trained K-Means model.
- `clustering_results.csv` – customer-level clustering results.
- `cluster_profile_summary.csv` – summary of customer characteristics by cluster.
- `elbow_method.png` – Elbow Method visualisation.
- `silhouette_scores.png` – silhouette analysis visualisation.
- `cluster_visualisation_tenure_charges.png` – visualisation of the resulting customer clusters.

All files are included in the **`Clustering_Analysis`** folder.

10. Conclusion

K-Means clustering was successfully applied to the prepared dataset containing **6,741 customer records and 10 features**.

The Elbow Method was used to evaluate different values of K and indicated a meaningful bend at approximately **K=4**. Four clusters were therefore selected for the final customer segmentation.

The final model identified four distinct customer groups with different patterns in tenure, monthly charges and observed churn outcomes. Cluster 3 showed the highest observed churn rate at **48.74%**, while Cluster 2 showed the lowest at **4.81%**.

The clustering results provide interpretable customer segments that can support further churn analysis and the development of targeted customer retention strategies.

11. Limitations

K-Means clustering assumes relatively compact groups and uses distance-based calculations when assigning observations to clusters.

The dataset also contains a combination of standardised continuous variables and encoded binary variables, which should be considered when interpreting the resulting segments.

Additionally, the cluster visualisation uses only `tenure` and `MonthlyCharges` for a two-dimensional representation, while the actual K-Means model was trained using all 10 features.

Finally, the cluster labels and business recommendations are descriptive interpretations of the resulting groups. They do not establish causal relationships between customer characteristics and churn.